

Entropy and second law [section 8.1.3]

for an adiabatic process:

$$s_2 - s_1 = \int \frac{dQ}{T} + s_{gen} \quad (8.17)$$

$$ds \geq 0$$

First law: you cannot win

Second law: you always lose

Calculation of entropy (for a reversible process)

$$\delta q = Tds \Rightarrow Tds - pdv = du_e$$

$$\Rightarrow Tds = du_e + pdv \quad (8.18)$$

$$Tds = du_e + pdv$$

use enthalpy

$$h = u_e + \underbrace{pv}$$

enthalpy → internal energy → work →

$$\Rightarrow dh = du_e + pdv + vdp$$

$$\Rightarrow \boxed{Tds = dh - vdp}$$

The above equation and 8.18 are alternative forms of 1st law

for a thermally perfect gas: $\underline{\underline{dh = c_p dT}}$ (8.10)

$$\Rightarrow \boxed{ds = c_p \frac{dT}{T} - \frac{vdp}{T}}$$

Entropy diff. Eqⁿ

assume:

$pv = RT$ (ideal gas law)

$$v = \frac{r}{M}$$

$$\Rightarrow \underline{ds} = c_p \frac{dT}{T} - R \frac{dp}{p} \Rightarrow \boxed{s_2 - s_1 = \int_{T_1}^{T_2} c_p \frac{dT}{T} - R \ln \frac{p_2}{p_1}} \quad (1)$$

for a calorically perfect gas ($c_p = \text{constant}$)

$$\left\{ \begin{array}{l} c_p = \text{cte} \\ \text{If we keep } c_p \text{ variable, } c_p = c_p(T) \text{ use Eq. } (1) \end{array} \right. \Rightarrow \boxed{s_2 - s_1 = c_p \ln \frac{T_2}{T_1} - R \ln \frac{p_2}{p_1}} \quad (2) \quad (\text{equivalent to 8.19b})$$

similarly: (use $p v = RT$, $c_p = \frac{\gamma R}{\gamma - 1}$; $c_v = \frac{R}{\gamma - 1}$ 8.16)

$$\Rightarrow \boxed{s_2 - s_1 = c_v \ln \frac{T_2}{T_1} + R \ln \frac{v_2}{v_1}} \quad (3) \quad \nu = \frac{1}{\rho} \quad \gamma = \frac{v}{u} = \frac{1}{\beta} \quad (\text{equivalent to 8.19c})$$

useful eqs to evaluate entropy increases

Isentropic relations

put $s_1 = s_2$ in (8.19a)

$$\Rightarrow \underline{0} = c_p \ln \frac{T_2}{T_1} - R \ln \frac{p_2}{p_1} \Rightarrow \ln \frac{p_2}{p_1} = \frac{c_p}{R} \ln \frac{T_2}{T_1}$$

$$\Rightarrow \frac{p_2}{p_1} = \left(\frac{T_2}{T_1} \right)^{c_p/R}$$

from (8.16)

$$\boxed{\frac{c_p}{R} = \frac{\gamma}{\gamma - 1}}$$

$$\Rightarrow \frac{p_2}{p_1} = \left(\frac{T_2}{T_1} \right)^{\gamma/(\gamma-1)}$$

$$\textcircled{2} \frac{p_2}{p_1} = \left(\frac{T_2}{T_1} \right)^{\frac{\gamma}{\gamma-1}}$$

"isentropic relation"

$$\textcircled{1} p = \gamma R T : \text{E. 9.5}$$

similarly:

$$\Rightarrow \frac{\rho_2}{\rho_1} = \left(\frac{T_2}{T_1} \right)^{1/(\gamma-1)}$$

in summary:

$$\boxed{\frac{p_2}{p_1} = \left(\frac{\rho_2}{\rho_1} \right)^\gamma = \left(\frac{T_2}{T_1} \right)^{\gamma/(\gamma-1)}} \quad \text{isentropic relations} \quad (*)$$

or:

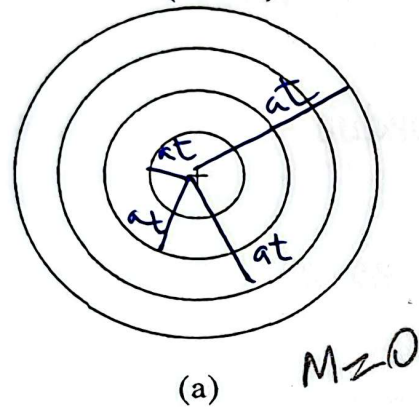
(8.20)

$$\frac{p}{\rho^\gamma} = \text{constant}; \quad \frac{T^{\gamma/(\gamma-1)}}{p} = \text{constant}; \quad \frac{T}{\rho^{(\gamma-1)}} = T v^{(\gamma-1)} = \text{constant}$$

(energy relation for an isentropic process) many processes can be assumed isentropic.

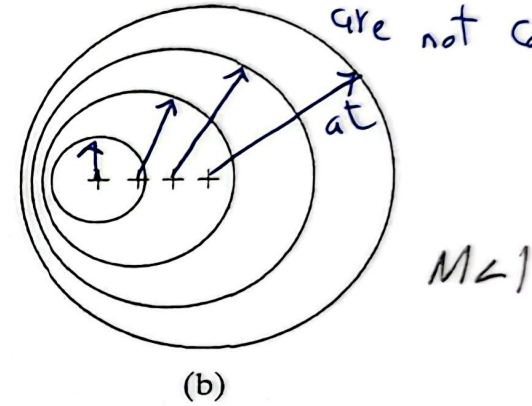
Speed of sound (8.1.4)

If I stand
spheres are
concentric



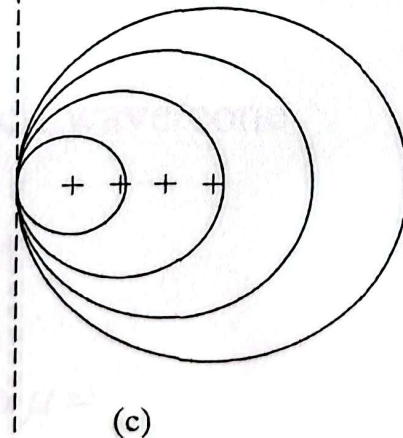
(a)

If I move, spheres
are not concentric



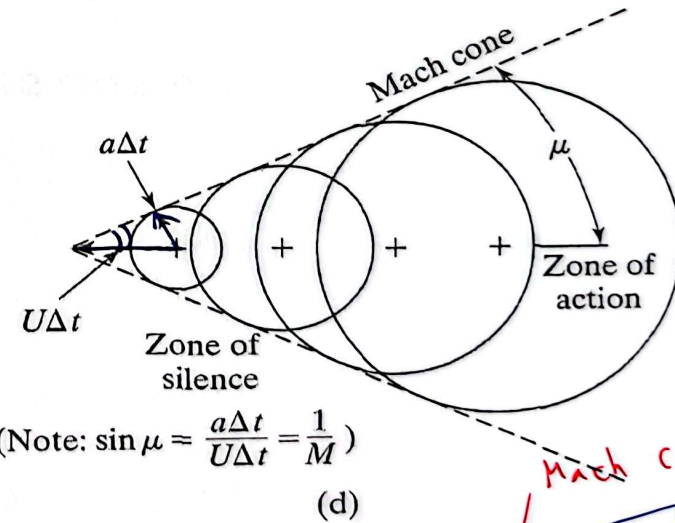
(b)

"Mach cone"
Zone of silence



(c)

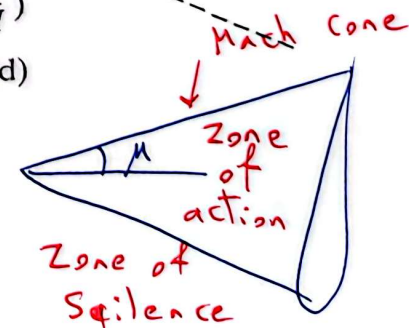
$M=1$



(d)

$M > 1$

$$\mu = \sin^{-1} \left(\frac{1}{M} \right)$$



(a) no external velocity

(b) Velocity $U < a$ - subsonic speed source stays inside family of circular sound waves ($Ut < at$)

(d) Velocity $U > a$ - supersonic speed source outside family of circles wave fronts form a disturbance envelope

$\Rightarrow$ Mach wave/cone

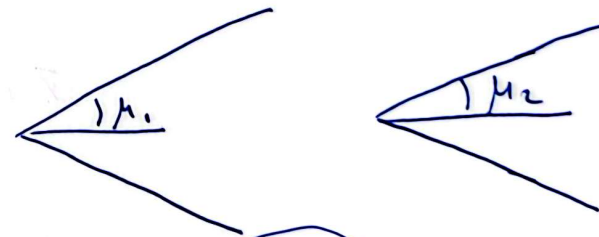
Mach angle:

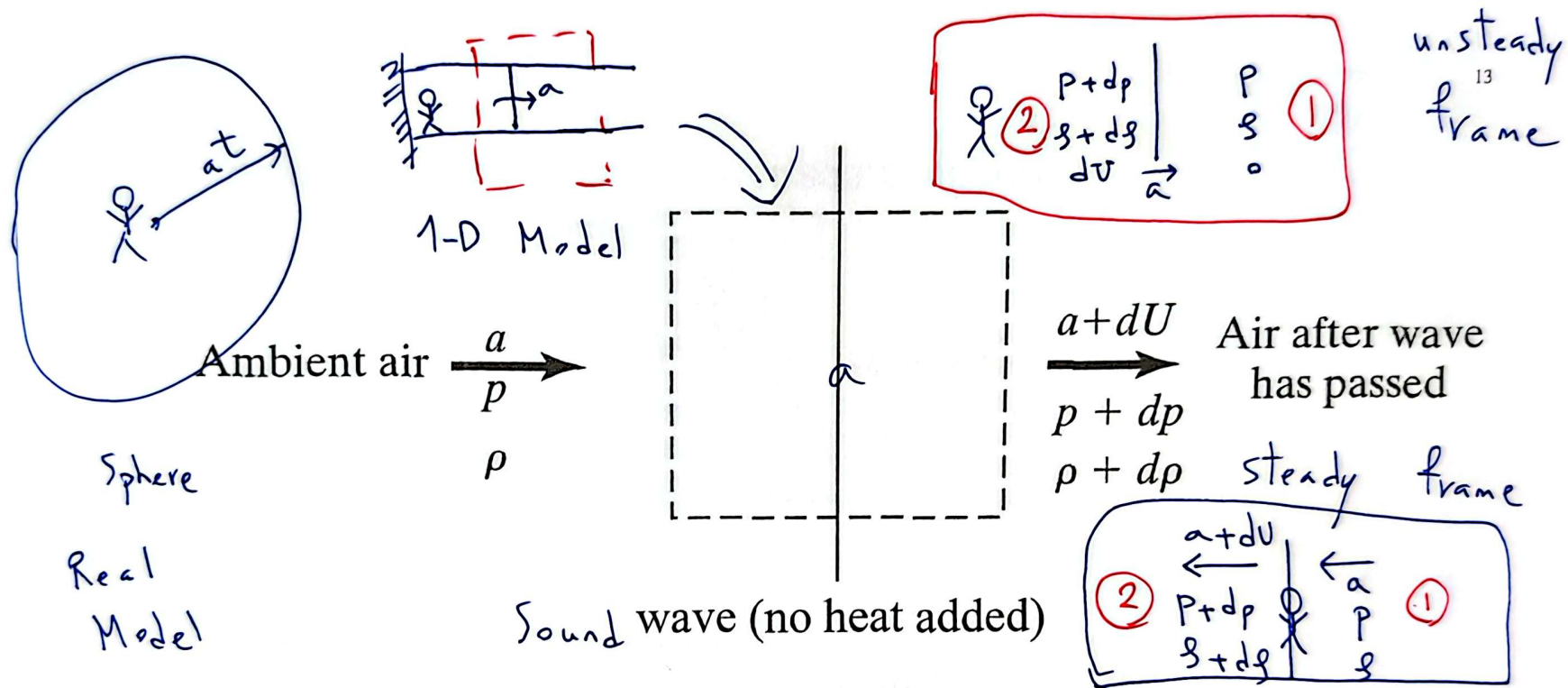
$$\sin \mu = \frac{at}{Ut} = \frac{a}{U} = \frac{1}{M}$$

$\Rightarrow$

$$\mu = \sin^{-1} \left(\frac{1}{M} \right) \quad (8.22)$$

$$M_1 > M_2 \rightarrow \mu_1 < \mu_2$$





assume a weak wave with speed a ; changes to speed, pressure

density, temperature are small $\Rightarrow$ isentropic

continuity: $\textcircled{1} \rho a = (\rho + d\rho) \textcircled{2} (a + dU)$ (8.23)

momentum: $\textcircled{1} p + \rho a^2 = (p + dp) + \textcircled{2} (\rho + d\rho)(a + dU)^2$ (8.24)

" $s a (a + dU)$ from ①

ignoring 2nd order terms, after some algebra

$$a = \sqrt{\left(\frac{\partial p}{\partial \rho}\right)_s}$$

$$\Rightarrow a = \sqrt{\left(\frac{\partial p}{\partial \rho}\right)_s} \quad \text{isentropic} \quad (8.26)$$

for a calorically perfect gas:

$$\underline{p} = \underline{\rho}^\gamma \times \underline{\text{constant}}$$

$$\frac{p}{\rho^\gamma} = \text{cte}$$

$$\Rightarrow a = \sqrt{\frac{\gamma p}{\rho}} = \sqrt{\gamma R T} \quad (8.28)$$

for standard sea level: $a_s = 340.9 \text{ m/s} = 1117 \text{ ft/s}$

$$M \equiv \frac{U}{a}$$

Mach number

$M < 1$ subsonic

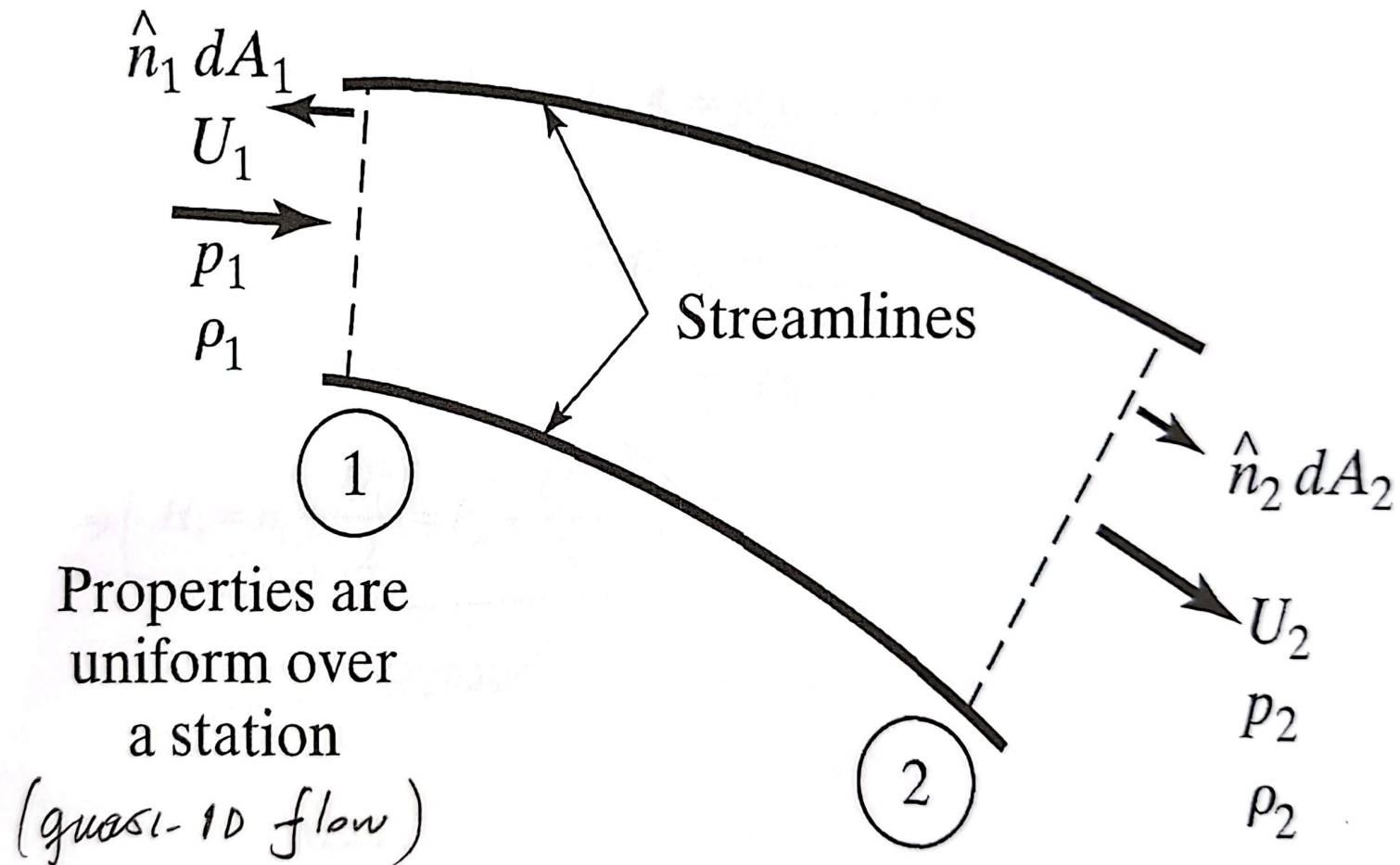
$M = 1$ sonic

$M > 1$ supersonic

for incompressible flow:

$$M = 0 \quad a \rightarrow \infty$$

Adiabatic Flow in a variable Area Streamtube [8.2]



Energy equation:

$$p_1 U_1 A_1 + u_{e1} \rho_1 U_1 A_1 + \frac{U_1^2}{2} \rho_1 U_1 A_1 = p_2 U_2 A_2 + u_{e2} \rho_2 U_2 A_2 + \frac{U_2^2}{2} \rho_2 U_2 A_2$$

Using continuity

$$\rho_1 U_1 A_1 = \rho_2 U_2 A_2$$

and enthalpy definition

$$h = u_e + p v \quad (v = 1/\rho)$$

$$\Rightarrow H_t = h_1 + \frac{U_1^2}{2} = h_2 + \frac{U_2^2}{2} \quad (8.29)$$

$H_t \equiv$ total (stagnation) enthalpy

Continuity: $\rho_1 u_1 = \rho_2 u_2$

Momentum: $p_1 + \rho_1 u_1^2 = p_2 + \rho_2 u_2^2$

Energy: $h_1 + \frac{u_1^2}{2} = h_2 + \frac{u_2^2}{2}$

For a perfect gas:

$$h + \frac{u^2}{2} = h_t = c_p T_t = \text{cte}$$

$\swarrow$ enthalpy $\downarrow$ k.E "if flow is adiabatic"

$$c_p T_1 + \frac{U_1^2}{2} = c_p T_2 + \frac{U_2^2}{2} \quad \cancel{c_p T_{t1} = c_p T_{t2}} \quad (8.30)$$

$$T_1 + \frac{U_1^2}{2c_p} = T_2 + \frac{U_2^2}{2c_p} \quad \boxed{T_{t1} = T_{t2}} \quad (8.31)$$

$$T_t = T + \frac{U^2}{2c_p} = T_{\text{static}} + T_{\text{dynamic}} = \text{constant} \quad (8.32)$$

$$T_{t1} = T_{t2} \quad (8.33)$$

(adiabatic process)

$$\underbrace{c_p T}_h + \frac{u^2}{2} = \underbrace{c_p T}_h$$

$$\xrightarrow{\div c_p} T + \frac{u^2}{2 c_p} = T_t$$

$$\xrightarrow{\div T} 1 + \frac{u^2}{2 c_p T} = \frac{T_t}{T}$$

$$\Rightarrow \frac{T_t}{T} = 1 + \frac{u^2}{2 \cancel{\frac{\gamma R T}{\gamma - 1}}} = 1 + \frac{u^2}{2 \cancel{a^2}} = 1 + \frac{\gamma - 1}{2} M^2$$

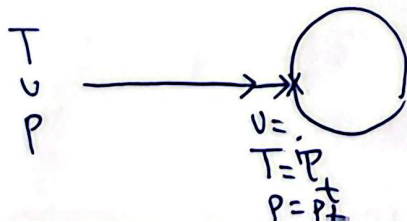
γ
" "
 c_p

$$\Rightarrow \boxed{\frac{T_t}{T} = 1 + \frac{\gamma - 1}{2} M^2}$$

① ✓

We know that $\frac{P_t}{P} = \left(\frac{T_t}{T} \right)^{\frac{\gamma}{\gamma - 1}} = \left(1 + \frac{\gamma - 1}{2} M^2 \right)^{\frac{\gamma}{\gamma - 1}} = \frac{P_t}{P}$ ② ✓

$$\boxed{\left(1 + \frac{\gamma - 1}{2} M^2 \right)^{\frac{1}{\gamma - 1}} = \frac{P_t}{P}} \quad ③$$



From (8.32) using equation 8.16 for c_p

total temp. $\leftarrow \frac{T_t}{T} = 1 + \frac{U^2}{2\gamma RT / (\gamma - 1)} = 1 + \frac{\gamma - 1}{2} M^2$ $\overline{T}_t = \overline{T}_o$ (8.34)

static temp. $\leftarrow T$

[Can be also derived starting momentum equation for isentropic inviscid flow-see text]

For isentropic flow: $\frac{p_t}{p} = \left(1 + \frac{\gamma - 1}{2} M^2\right)^{\gamma/(\gamma - 1)}$ $\overline{p}_t = \overline{p}_o$ (8.36)

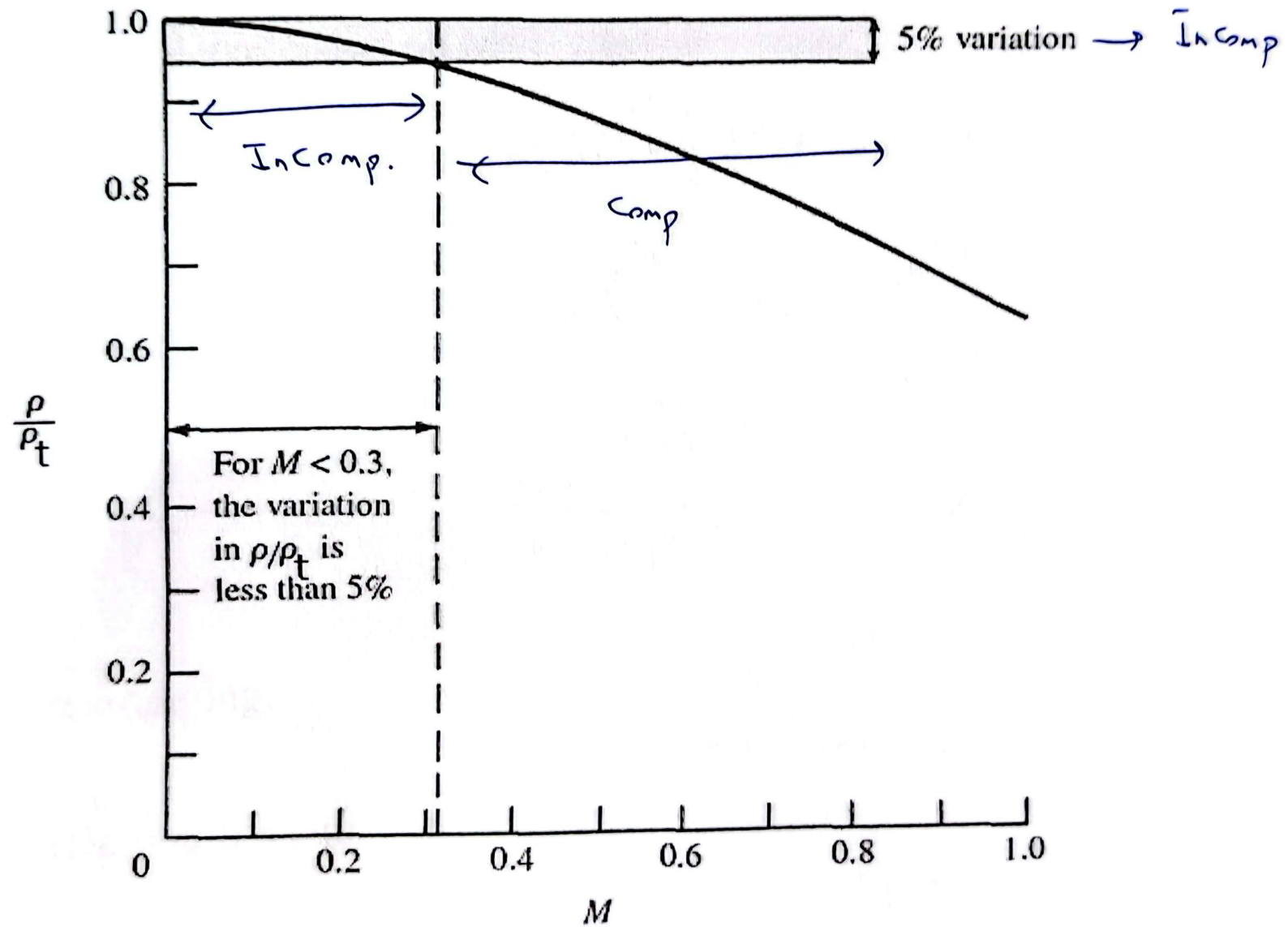
(using 8.20) $\leftarrow p$ static pressure $\frac{\rho_t}{\rho} = \left(1 + \frac{\gamma - 1}{2} M^2\right)^{1/(\gamma - 1)}$ $\overline{\rho}_t = \overline{\rho}_o$ (8.37)

p_t and ρ_t are stagnation (total) pressure and density

from (8.36) 5% error in density for $M = 0.32$ 1% for $M = 0.14$
10% for $M = 0.44$

8.34, 8.36, 8.37 are tabulated in Table 8.1 (isentropic tables)

If $M = 0.32 \rightarrow$ 8.37 is 1.05
we assume $\rho = \rho_o$ if $M < 0.32$



$$\frac{p_t}{p} = \left(1 + \frac{\gamma-1}{2} M^2\right)^{\frac{\gamma}{\gamma-1}}, \quad \frac{p}{p_t} = \left(1 + \frac{\gamma-1}{2} M^2\right)^{-\frac{\gamma}{\gamma-1}}$$

binomial expansion on M^2 :

(for small M^2)

$$\frac{p_t}{p} = \left(1 + \frac{\gamma}{2} M^2 + \frac{\gamma}{8} M^4 + \dots\right) \quad (8.38)$$

for a low M :

$$\Rightarrow \boxed{\frac{p_t}{p} = \left(1 + \frac{\gamma}{2} M^2\right)} \quad (8.39)$$

for a perfect gas:

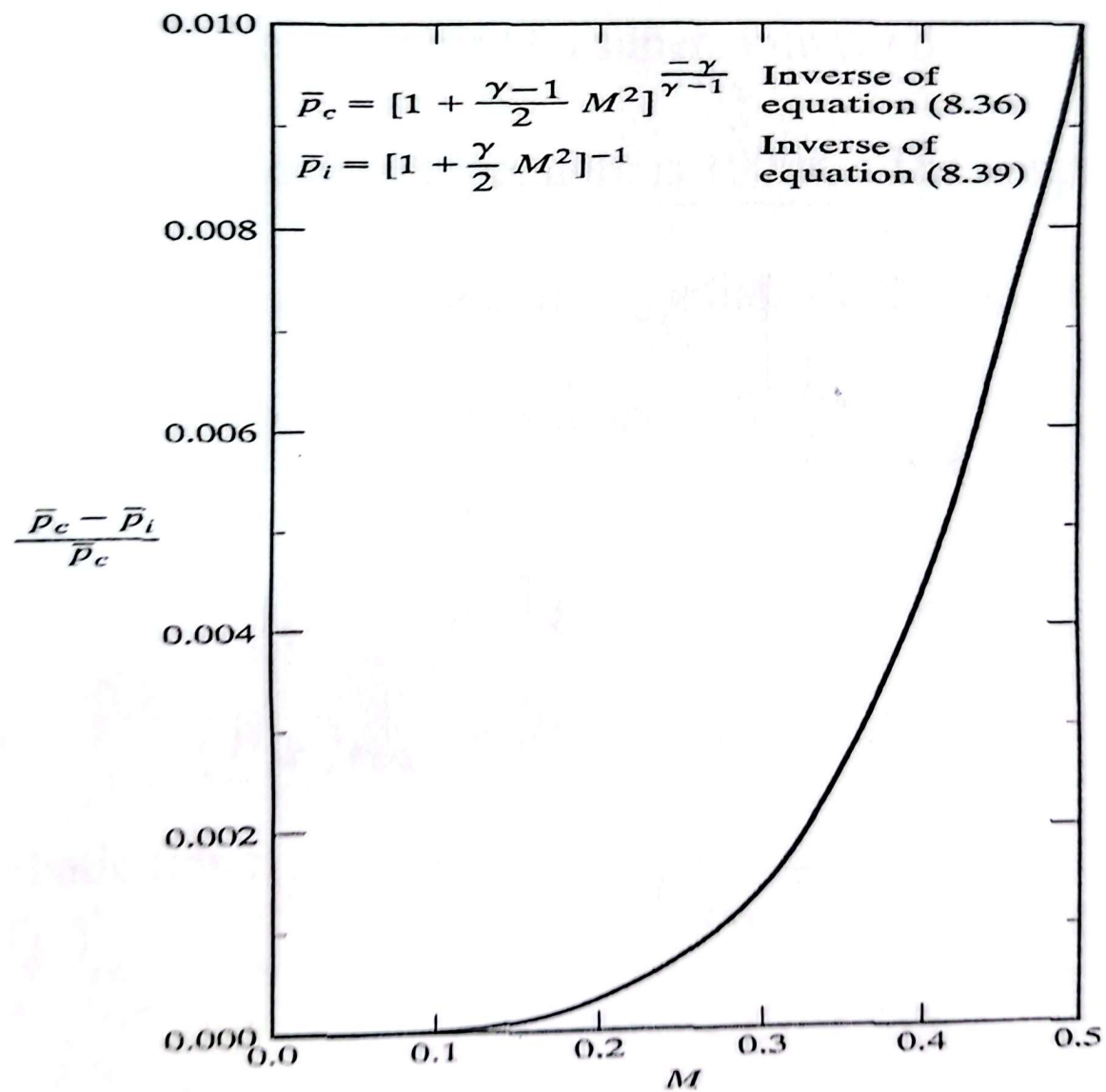
$$\frac{p_t}{p} = 1 + \frac{\gamma U^2}{2\gamma RT} = 1 + \frac{U^2}{2p/\rho} \quad (8.40)$$

rearranging:

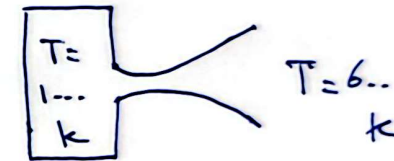
$$\boxed{p_t = p + \frac{\rho U^2}{2} \left(1 + \frac{\gamma}{8} M^4 + \dots\right)} \quad (8.41)$$

(Bernoulli's equation)

$$\left\{ \begin{array}{l} \text{error} \approx 1\% \text{ for } M = 0.5 \\ 0.15\% \text{ for } M = 0.3 \end{array} \right.$$



Example: In the reservoir of a supersonic wind tunnel, the velocity is negligible, and the temperature is 1000K. The temperature at the nozzle exit is 600K. Assuming adiabatic flow through the nozzle, calculate the velocity at the exit:



$U=0$
(total conditions)

Reservoir:

$$H_t = C_p T_t$$

Exit:

$$(H_t)_{res} = (H_t)_{exit} = C_p T_e + \frac{V_e^2}{2}$$

Adiabatic flow:

$$\Rightarrow C_p (T_t)_{res} = C_p T_e + \frac{V_e^2}{2}$$

$$(T_t)_{res} = (T_t)_{exit}$$

$$C_{p,air} = \frac{\gamma R}{\gamma - 1} = \frac{1.4}{1.4 - 1} \times 287 = 1004.5 \frac{\text{Joules}}{\text{kg}^\circ\text{K}}$$

$$\Rightarrow V_e = \sqrt{2 C_p ((T_t)_{res} - T_e)}$$

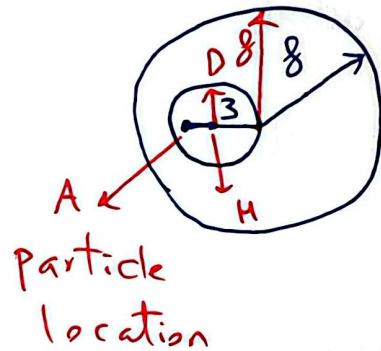
$$= 896.4 \text{ m/s}$$

$$M = \sqrt{3}$$

$$U = M \sqrt{\gamma R T}$$

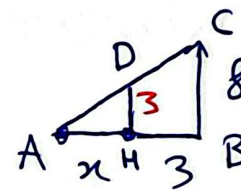
$$\Rightarrow U = \frac{1.73}{1.4} \sqrt{1.4 \times 287 \times 600} = 1.73$$

Ex.1) A particle moving with constant speed creates two spherical disturbances as shown in the figure below. The radius of first sphere is 8 meters and the radius of the second sphere is 3 meters. Find the Mach number of the particle.



Solution:

we create two triangles



$$\triangle ABC \sim \triangle ADH$$

$$\frac{AH}{DH} = \frac{AB}{BC}$$

$$\frac{x}{3} = \frac{x+3}{8}$$

$$\Rightarrow x = \frac{9}{5} = 1.8$$

$$M = \frac{AH}{DH} = \frac{1.8}{3} = 0.6$$

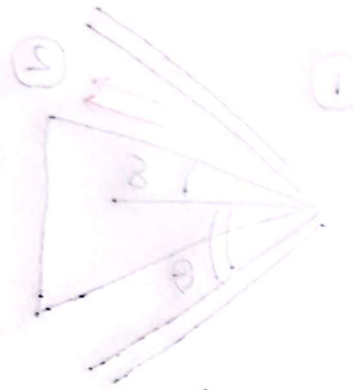
$$8x = 3x + 9$$

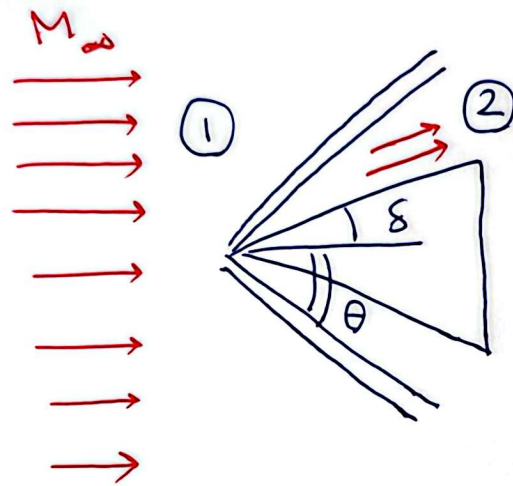
$$5x = 9 \rightarrow x = \frac{9}{5} = 1.8$$

Normal and Oblique Shocks [8.6-8.10]

Objectives:

- Develop relations for normal and oblique shocks
- Study viscous effects, shock boundary layer interactions
- Introduction to shock/shock interactions
- Introduction to supersonic tunnel testing

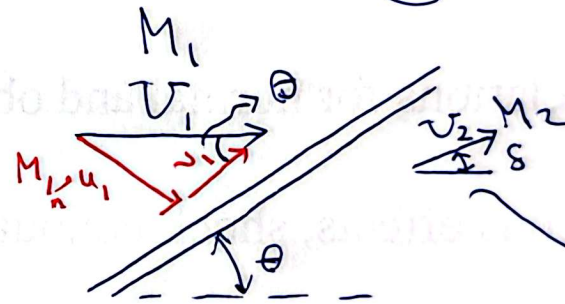




oblique shock rotates the flow
s. that flow becomes parallel
to wedge.

$$V_2 < V_1$$

$$M_2 < M_1$$



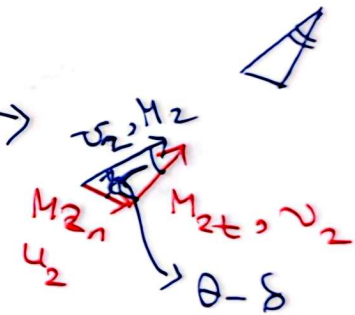
$\theta \equiv \beta$: shock angle

$\delta \equiv \theta$: wedge angle
(flow deflection
angle)

$$\begin{cases} u_1 = V_1 \sin \theta \\ M_{1n} = M_1 \sin \theta \end{cases}$$

normal
Mach

$$\begin{cases} M_{1t} = M_1 \cos \theta \\ V_1 = V_1 \cos \theta \end{cases}$$



$$\begin{cases} u_2 = V_2 \sin(\theta - \delta) \\ M_{2n} = M_2 \sin(\theta - \delta) \\ M_{2t} = M_2 \cos(\theta - \delta) \\ V_2 = V_2 \cos(\theta - \delta) \end{cases}$$